

BookRec

A Goodreads Book Recommender

Exploratory data analysis → collaborative filtering vs. a popularity baseline → an LLM re-ranking layer that personalizes the Top-N.

EXECUTIVE SUMMARY

A sparse, blockbuster-driven, positively-skewed catalog makes a strong baseline hard to beat — so we pair collaborative filtering with an AI re-ranking layer.

1 98.5%
matrix sparsity

1,192 readers × 9,229 books — collaborative filtering must work from very little overlap.

2 6%
top 1% of books

A blockbuster long tail (Gini 0.60). We report catalog coverage, not just accuracy.

3 65%
ratings are 4–5 stars

Mean 3.84/5. RMSE rewards predicting “high,” so a popularity baseline is a serious benchmark.

4 735
books never rated

Plus 54% with <10 ratings. Cold-start is an item problem; user cold-start is ~0 by construction.

RECOMMENDED APPROACH

UBCF generates candidates · a popularity/bias baseline is the guardrail & cold-start fallback · an LLM layer adds explainable personalization — graduate from offline P/R@K to A/B-tested online lift.

The dataset is small, dense-by-construction, and highly concentrated.

PROJECT 2 · DATA AT A GLANCE

A sparse, positively-skewed, blockbuster-driven catalog

Six numbers that frame every modeling decision that follows.

164,728

Ratings in sample

1,192

Unique readers

9,964

Books in catalog

1.497%

User-item matrix density

3.84 / 5

Mean rating · 65% are 4-5 stars

54%

Books with <10 ratings (item cold start)

Source: Project 2 ratings sample (Goodreads-derived) · loaded via bookrec.data_loader

A few blockbusters dominate — the top 1% of books capture 6% of all ratings.

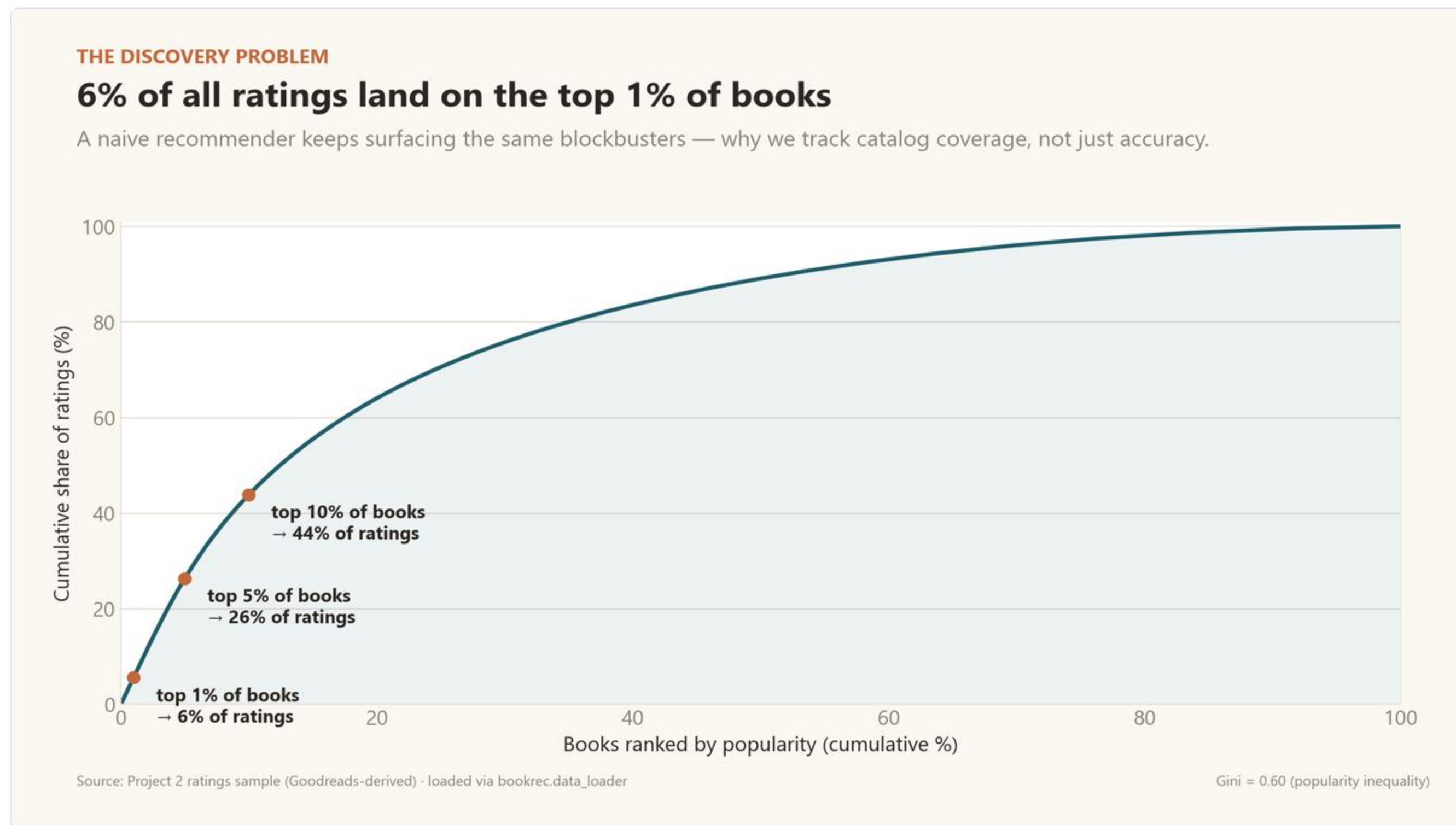


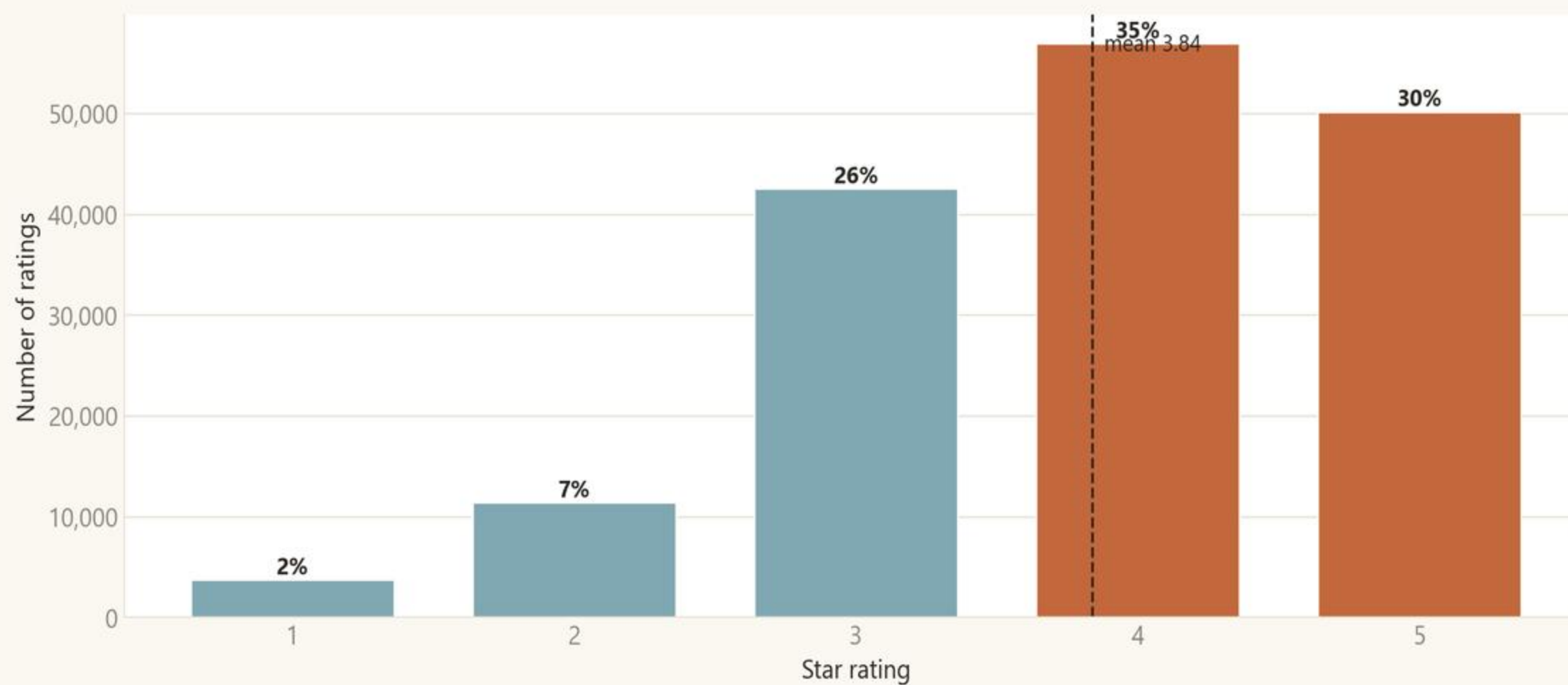
EXHIBIT 3 · FEEDBACK IS NOT NEUTRAL

65% of ratings are 4–5★, so a popularity baseline is genuinely hard to beat.

RATINGS ARE NOT NEUTRAL FEEDBACK

65% of ratings are 4 or 5 stars

People rate what they already expected to like (missing-not-at-random). RMSE rewards “predict high” — so a popularity baseline is a serious benchmark.



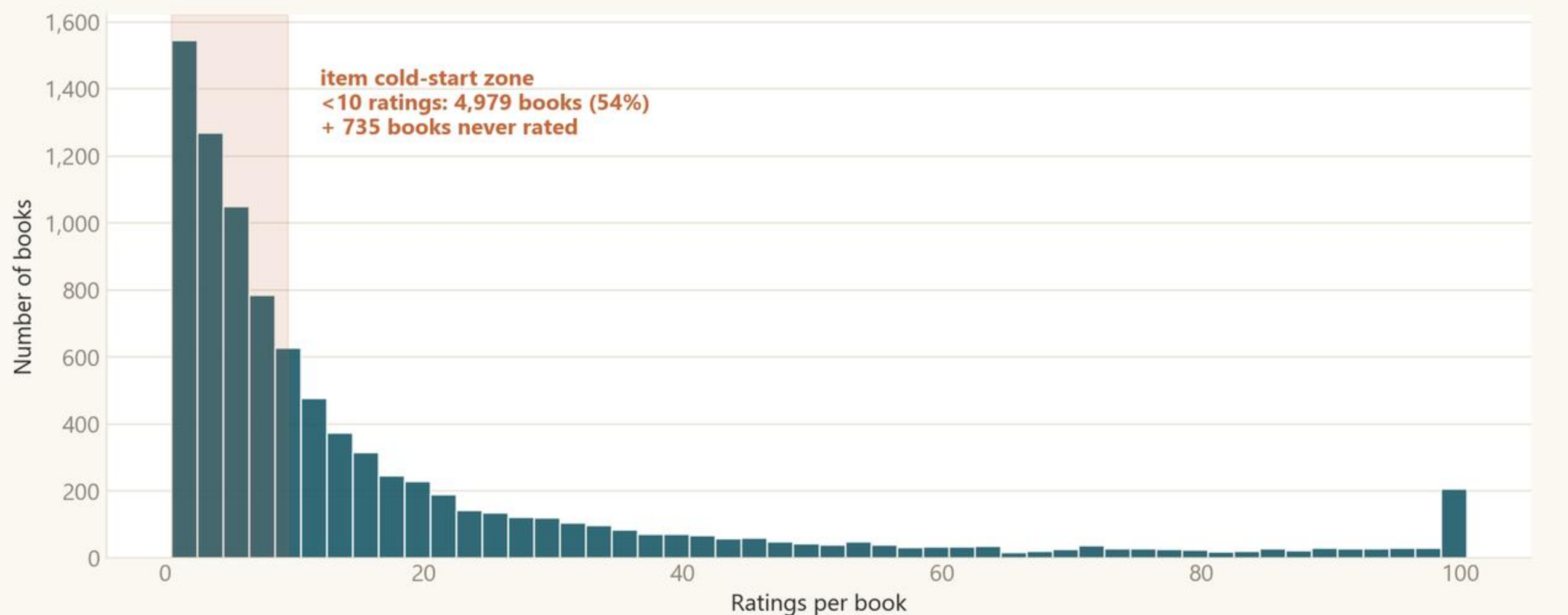
Source: Project 2 ratings sample (Goodreads-derived) · loaded via bookrec.data_loader

The cold-start risk sits on the books, not the readers.

THE COLD-START RISK IS ON THE BOOKS, NOT THE READERS

4,979 of 9,229 rated books have fewer than 10 ratings

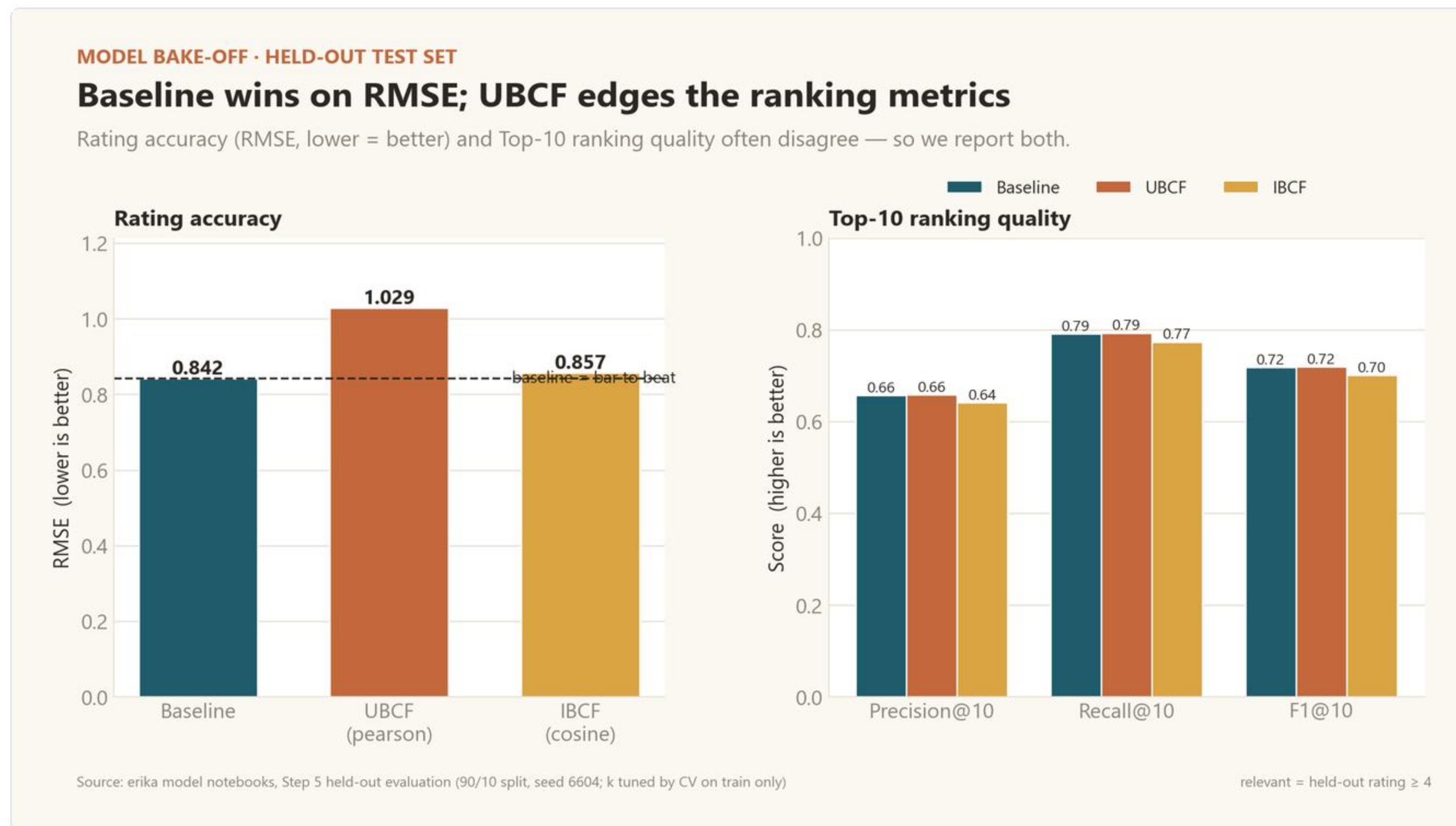
Every reader here has 100+ ratings (a filtered core), so thin signal lives on the catalog side — plus 735 books with no ratings at all.



Source: Project 2 ratings sample (Goodreads-derived) · loaded via bookrec.data_loader

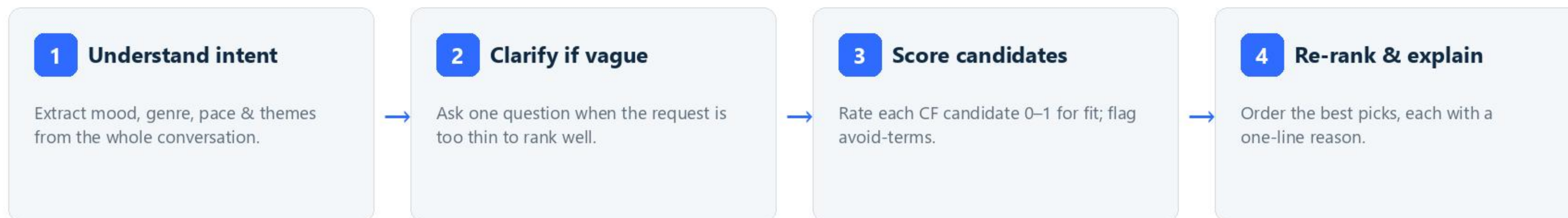
Median 8 · mean 18 ratings per book

UBCF wins the Top-N ranking metrics; the baseline wins RMSE — so we optimize for ranking.



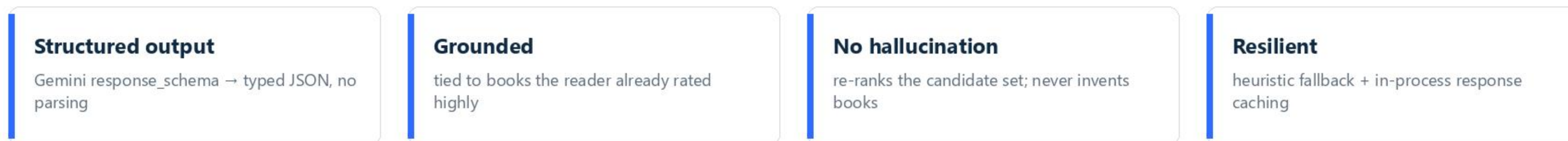
An LLM re-ranks — never invents — the CF Top-N, grounded and explained.

FROM THE CF TOP-N (UBCF)



... TO A PERSONALIZED, EXPLAINED TOP-N SHOWN IN CHAT

DESIGN PRINCIPLES



MODEL & SAFETY

Google Gemini 2.5-flash-lite · API key read from the environment, never committed · falls back to a transparent heuristic when no key is set.

Collaborative filtering drives discovery; the LLM layer adds explainable personalization.

Collaborative filtering (UBCF / IBCF)

APPLICATIONS

- “Readers like you also enjoyed...” discovery, cross-sell & retention
- Cheap to serve once trained; needs only behavior, no metadata

CHALLENGES

- Cold-start for new users/books; data sparsity; popularity bias
- Offline metrics (RMSE, P/R@K) don’t guarantee online lift

LLM re-ranking layer

APPLICATIONS

- Mood- & intent-aware personalization with a reason per pick
- Conversational discovery and merchandising; builds user trust

CHALLENGES

- API cost & latency at scale; rate limits
- Grounding & safety — must re-rank real candidates, never invent

RECOMMENDED APPROACH

UBCF candidates + a popularity/bias baseline as guardrail + the LLM re-ranker for explainable personalization; cache LLM calls to control cost; graduate from offline P/R@K to A/B-tested online lift.

■ FOR REFERENCE

Technical Appendix

Nine detailed exploratory exhibits (A1–A9): matrix sparsity, concentration & Gini, Zipf fit, bias decomposition, Bayesian shrinkage, engagement, vintage, metadata completeness, and the data-quality audit.